# Experiment Title: File and Directory Management in Linux Terminal

**Topic:** Creating, Copying, Moving, and Removing Files and Directories

## 1. Objective

To practice Linux file and directory management commands:

- Create folders and nested paths (**`mkdir`**, **`mkdir -p`**).
- Create empty files (**`touch`**).
- Copy, move, and rename files (**`cp`**, **`mv`**).
- Delete files and directories (**`rm`**, **`rm -r`**, **`rm -rf`**).

## 2. Theoretical Background

Linux provides commands to create, duplicate, move, and delete files directly from the terminal without a GUI.

### 2.1 Commands Practiced in This Lab

| Command | Syntax | Simple Meaning / Purpose | Example |
|---|---|---|---|
| `mkdir` | `mkdir <name>` | Create new directory. | `mkdir docs` |
| `mkdir -p` | `mkdir -p <path>` | Create parent and child folders together. | `mkdir -p proj/src/utils` |
| `touch` | `touch <file>` | Create empty file or update timestamp. | `touch notes.txt` |
| `cp` | `cp <src> <dest>` | Copy file to new location/name. | `cp notes.txt backup.txt` |
| `mv` | `mv <src> <dest>` | Move or rename file/folder. | `mv old.txt new.txt` |
| `rm` | `rm <file>` | Delete file permanently. | `rm temp.log` |
| `rm -r` | `rm -r <folder>` | Delete folder and its contents recursively. | `rm -r old_data` |
| `rm -rf` | `rm -rf <path>` | Force delete folder and files without prompt. | `rm -rf build` |

### 2.2 Important Rules for File Operations

- **No Trash Bin:** `rm` deletes files immediately and permanently.
- **Parent Flag (`-p`):** Creates missing parent folders automatically.
- **Recursive Flag (`-r`):** Removes non-empty folders and their contents.
- **Force Flag (`-f`):** Ignores errors/prompts, deleting without warning.

## 3. Equipment / Software Requirements

| Item | Details | Item | Details |
|---|---|---|---|
| **Operating System** | Linux (Ubuntu 24.04 LTS) | **Host Machine** | `firefly` |
| **User Account** | `ratul` | **Command Shell** | GNU Bash 5.2 / Zsh |
| **Terminal Emulator** | Linux Terminal (Kitty / Ghostty) | **Core Tools** | GNU Coreutils (`mkdir`, `touch`, `cp`, `mv`, `rm`) |

## 4. Procedure / Steps

Steps followed in this experiment:

1. Create demo folder (**`mkdir lab3_demo`**) and navigate inside (**`cd`**).
2. Create folder **`docs`** (**`mkdir`**) and nested path **`proj/src/utils`** (**`mkdir -p`**).
3. Create empty files **`notes.txt`** and **`main.c`** using **`touch`**.
4. Copy **`notes.txt`** to **`notes_copy.txt`** using **`cp`**.
5. Move copy to **`backup/`** and rename to **`notes_v1.txt`** using **`mv`**.
6. Delete file with **`rm`**, and delete directories with **`rm -r`** and **`rm -rf`**.

## 5. Result / Output Demonstration

All commands were executed under user account `ratul` on host machine `firefly`.

### 5.1 Session 1: Creating Directories, Nested Trees, Empty Files, and Copying

ratul@firefly: ~/lab3_demo — bash

```
ratul@firefly:~/lab3_demo$ mkdir docs backup
ratul@firefly:~/lab3_demo$ mkdir -p proj/src/utils
ratul@firefly:~/lab3_demo$ touch docs/notes.txt proj/src/main.c
ratul@firefly:~/lab3_demo$ ls docs proj/src
docs:  notes.txt        proj/src:  main.c  utils
ratul@firefly:~/lab3_demo$ cp docs/notes.txt docs/notes_copy.txt
ratul@firefly:~/lab3_demo$ ls -l docs
total 0
-rw-r--r- 1 ratul users 0 Oct  8 14:10 notes.txt
-rw-r--r- 1 ratul users 0 Oct  8 14:12 notes_copy.txt
ratul@firefly:~/lab3_demo$ _
```

**Explanation of Session 1 Results:**

- `mkdir` and `mkdir -p`: Created folders; `-p` created the entire path `proj/src/utils` at once.
- `touch`: Created empty 0-byte files `notes.txt` and `main.c`.
- `cp`: Made an exact copy of `notes.txt` as `notes_copy.txt`.

### 5.2 Session 2: Moving, Renaming, and Removing Files and Directories

ratul@firefly: ~/lab3_demo — bash

```
ratul@firefly:~/lab3_demo$ mv docs/notes_copy.txt backup/
ratul@firefly:~/lab3_demo$ mv backup/notes_copy.txt backup/notes_v1.txt
ratul@firefly:~/lab3_demo$ ls backup
notes_v1.txt
ratul@firefly:~/lab3_demo$ touch temp.log && rm temp.log
ratul@firefly:~/lab3_demo$ mkdir -p temp_cache/data && touch temp_cache/data/cache.bin
ratul@firefly:~/lab3_demo$ rm -r temp_cache
ratul@firefly:~/lab3_demo$ mkdir -p scratch/a/b && touch scratch/a/test.tmp
ratul@firefly:~/lab3_demo$ rm -rf scratch
ratul@firefly:~/lab3_demo$ ls
backup   docs   proj
ratul@firefly:~/lab3_demo$ _
```

**Explanation of Session 2 Results:**

- `mv`: Relocated `notes_copy.txt` into `backup/` and renamed to `notes_v1.txt`.
- `rm`: Deleted file `temp.log` permanently.
- `rm -r` and `rm -rf`: Recursively removed `temp_cache` and force-deleted `scratch` without prompt.
- Verified remaining folders with `ls`.

## 6. Conclusion

Key takeaways from this lab:

1. Created directories with `mkdir` and nested paths with `mkdir -p`.
2. Created files with `touch` and duplicated files with `cp`.
3. Moved and renamed files with `mv`.
4. Deleted files with `rm` and directory trees with `rm -r` and `rm -rf`.